depends on two things: The distance between the regions and the amount of time that has elapsed since the bang. If their separation is less than the distance light could have traveled in the time since the bang, then the regions could have influenced each other; otherwise, they couldn't have. Now, you might think that *all* regions of the observable universe could have interacted with each other way back near the beginning because the farther back we wind the film, the closer the regions become and hence the easier it is for them to interact. But this reasoning is too quick; it doesn't take account of the fact that not only were regions of space closer, but there was also less time for them to have communicated.

To do a proper analysis, imagine running the cosmic film in reverse while focusing on two regions of space currently on opposite sides of the observable universe—regions that are so distant that they are currently beyond each other's spheres of influence. If in order to halve their separation we have to roll the cosmic film more than halfway back toward the beginning, then even though the regions of space were closer together, communication between them was still impossible: they were half as far apart, but the time since the bang was *less* than half of what it is today, and so light could travel only *less* than half as far. Similarly, if from that point in the film we have to run more than halfway back to the beginning in order to halve the separation between the regions once again, communication becomes more difficult still. With this kind of cosmic evolution, even though regions were closer together in the past, it becomes more puzzling—not less—that they somehow managed to equalize their temperatures. Relative to how far light can travel, the regions become increasingly cut off as we examine them ever farther back in time.

This is exactly what happens in the standard big bang theory. In the standard big bang, gravity acts only as an attractive force, and so, ever since the beginning, it has been acting to slow the expansion of space. Now, if something is slowing down, it will take more time to cover a given distance. For instance, imagine that Secretariat left the gate at a blistering pace and covered the first half of a racecourse in two minutes, but because it's not his best day, he slows down considerably during the second half and takes three more minutes to finish. When viewing a film of the race in reverse, we'd have to roll the film more than halfway back in order to see Secretariat at the course's halfway mark (we'd have to run